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Master's Thesis

Methodological Reflection on Data-Driven Research Under Uncertainty:

An Empirical Study of Early User Feedback and Mobile Application Popularity

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Abstract

The original objective of this research was to empirically investigate whether first impressions—captured through early user sentiment and engagement signals—predict the long-term popularity of mobile applications. The study was initially designed and executed following a conventional, Waterfall-style research methodology: hypotheses, analytical methods, and evaluation criteria were specified upfront, and validation was deferred to the final stage. While this approach yielded technically correct analyses, it ultimately produced limited and marginal insights relative to the effort invested.

Rather than treating this outcome as a purely negative result, the work adopts a reflexive stance and reinterprets the study itself as a methodological use case. Through systematic self-criticism, the research examines how rigid evaluation practices, confirmation bias, and a fear of missing a “better” method or metric—conceptually aligned with Fear of Missing Out (FoMO)—contributed to delayed iteration, over-engineering, and reduced conceptual progress. This introspective analysis revealed that the core limitation was not the empirical question, but the research strategy used to pursue it.

Building on this diagnosis, the work synthesizes insights from evaluation theory, developmental evaluation, design science research, and Agile/DevOps practices to propose a Purpose-Oriented Benchmark Framework. The framework reconceptualizes evaluation as an iterative, learning-driven process, explicitly distinguishing between high-sensitivity formal metrics and lightweight, adaptive benchmarks suitable for exploratory empirical research. The original mobile-application study is then reused as a concrete demonstration of this framework, contrasting initial unsupervised clustering approaches with later decision-rule-based iterations to show how iterative benchmarking can expose unproductive paths, guide methodological pivots, and improve conceptual clarity.

By transforming a partially unsuccessful empirical study into a methodological case study, this work demonstrates how research failure can serve as a source of methodological insight rather than mere error. It contributes a human-centered perspective on research design, highlighting the interaction between researcher cognition and methodological structure, and lays the groundwork for future investigation into adaptive benchmarking, stopping rules for exploratory research, and the role of FoMO in professional research practice.

CHAPTER 1

Introduction

Data-driven research increasingly relies on large-scale observational datasets, complex analytical pipelines, and flexible modeling techniques. In many contemporary research settings, particularly those involving web data and unsupervised exploration, the availability of data and methods far exceeds the clarity of the empirical signal they contain.

This situation raises a fundamental methodological question: how should research be structured and governed when empirical structure is weak, ambiguous, or continuously distributed rather than discrete? Traditional research designs often assume that sufficient signal will emerge to guide successive analytical stages. When this assumption does not hold, research may remain technically rigorous while producing limited epistemic resolution.

This thesis addresses this problem through a dual approach. It combines a methodological analysis of research strategies with an empirical study conducted on mobile application data. The empirical study examines the relationship between early user feedback and long-term application popularity using large-scale data scraped from the Google Play Store. The methodological analysis examines how research processes evolve when confronted with empirical uncertainty.

The choice of mobile application data is motivated by its relevance to contemporary software ecosystems and by the richness of publicly available behavioral signals such as ratings, reviews, and update histories. Prior research has investigated release strategies and update practices in mobile application marketplaces, often emphasizing routine timing and developer-centric decision-making. However, the extent to which early user feedback provides actionable structure for guiding research and decision processes remains insufficiently understood.

Rather than positioning the empirical study solely as a hypothesis-testing exercise, this thesis treats it as a concrete instance of data-driven research conducted under realistic academic constraints. The empirical analysis follows established methodological practices, including feature engineering, exploratory analysis, and unsupervised modeling, thereby providing a representative research trajectory rather than an optimized experimental setup.

The central contribution of this work is methodological. By examining both the outcomes and the execution of the empirical study, the thesis investigates how research strategies interact with empirical ambiguity, how analytical progression is implicitly governed, and how behavioral and structural factors influence methodological decision-making.

To address these issues, the thesis is organized into two main components. The first develops a methodological perspective on research strategies, including linear and iterative designs, evaluation practices, and behavioral dimensions of research execution. The second presents the empirical study as a use case, documenting the analytical process and its outcomes without assuming conclusive results.

By linking these components through synthesis and reflection, the thesis aims to contribute to a broader understanding of how data-driven research can be designed, evaluated, and governed in contexts where empirical uncertainty is not an exception, but a defining condition.

PART I

Methodological perspective

CHAPTER 2

The Cognitive Methodologies of Research

2.1 Introduction: Methodology as a State of Mind

Methodologies in research are not only project management strategies but also reflections of cognitive states. The *Waterfall approach* represents a linear, control-oriented mindset seeking certainty and completeness. The *Agile approach*, in contrast, embodies adaptability, iteration, and comfort with uncertainty. This chapter explores how these two methodological paradigms parallel psychological behaviors in researchers, particularly the role of **FOMO (Fear of Missing Out)** as a driver of delay and over-planning [13, 39].

2.2 Fear of Missing Out in Research

2.2.1 Definition and Origin

The concept of FoMO, originally defined by Przybylski et al. [39], refers to the pervasive apprehension that others might be having experiences or acquiring knowledge from which one is absent. In research, this manifests as anxiety about missing new publications, methods, or datasets that might invalidate ongoing work [13].

2.2.2 FoMO as a Cognitive Bias in Research

Empirical studies [li2023fomo, zhang2024fomo, 27] link FoMO to procrastination, reduced self-control, and inefficient decision-making. For researchers, this translates into endless literature reviews, perfectionism, and delayed experimentation. FoMO thus functions as a psychological barrier to progress, reinforcing the desire for exhaustive completeness.

2.2.3 The Waterfall Mindset as a FoMO Response

The traditional *Waterfall* approach aligns with a FoMO-driven psychology: a belief that no step can begin until the previous one is fully understood and completed. This linear mindset provides an illusion of control that momentarily soothes FoMO anxiety, but it also traps researchers in a cycle of over-preparation and delayed execution [4, 11].

2.3 Waterfall as Cognitive Architecture

2.3.1 The Illusion of Completeness

The Waterfall model appeals to the desire for total clarity before action. Cognitively, this mirrors perfectionism and low ambiguity tolerance. Researchers experiencing FoMO often use exhaustive planning as a defense mechanism against uncertainty [13].

2.3.2 Confirmation and Sunk-Cost Bias

Once a research direction is chosen, confirmation bias reinforces adherence to initial hypotheses, while the sunk-cost effect discourages iteration or change [23]. This creates rigidity—an emotional form of commitment to the original plan, which mirrors the sequential rigidity of the Waterfall model.

2.3.3 Cognitive exhaustion trap:

"The more I learn, the more I realize how much I don't know" - Einstein (popularisation of Socrates' philosophy). In the researcher's mind, it's a constant doubt. The researcher is under pressure to showcase knowledge while simultaneously feeling humbled by the lack of it. New tools and datasets emerge faster than one can adapt, feeding the sense of always being behind. Researchers stuck in linear planning often experience stagnation while the field moves forward [aslam2024fomo]. They can only free themselves by embracing the void without the urge to fill it.

2.4 Case Study: The Waterfall Mindset in Practice

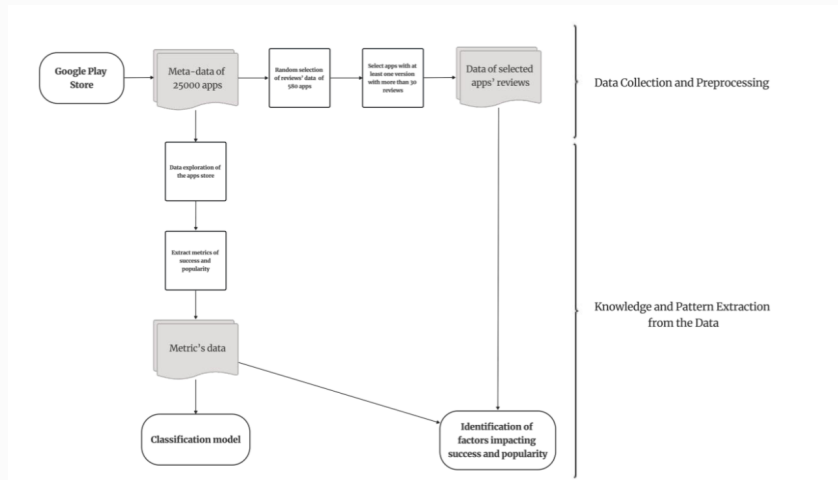
The following case study illustrates a real research process executed under a Waterfall mindset, demonstrating how the overconfidence in completeness and FoMO-driven design can lead to disproportionate effort and limited insight.

Case Study – Methodology: “The Impact of First Impressions on the Popularity of Mobile Applications”

Context and Objective. In this study we investigated how the *first impression* of users, expressed through early reviews on the Google Play Store, affects an application's long-term popularity. The goal was to evaluate whether early feedback could serve as a predictive factor for final success and to design models integrating user sentiment and update cycles into release strategies.

Research Questions.

1. RQ1: What is the impact of the first impression on user feedback on mobile apps?
2. RQ2: What are the key topics of the first impression identified using NLP?



General Design (Waterfall Sequence). The methodological design strictly followed a Waterfall model:

1. Literature synthesis and fixed hypothesis formation.
2. Data crawling and preprocessing pipeline (Python **Scrapy** framework).
3. Feature engineering and construction of compound popularity metrics.
4. Descriptive statistical analysis and correlation validation.
5. Classification and clustering modeling (supervised + unsupervised hybrid).
6. Evaluation and interpretation.

At each stage, validation depended on full completion of the prior one. No feedback or iteration loops were included, as I assumed the hypothesis would be confirmed.

Data Collection and Preparation.

Two datasets were built:

- *Meta-data dataset* – scraped app-level information (ratings, installs, categories, and update logs).
- *Review dataset* – containing user comments from initial and final versions.

Version selection was based on dispersion metrics (min/mean/max intersection) to capture the most representative “first impression” version. Data were cleaned and normalized manually for each variable before proceeding.

Feature Engineering. Descriptive analyses revealed the need for new synthetic variables combining metrics such as score, reviews, and version count. A feature representing “popularity evolution” was designed through correlation-based synthesis, serving as input to later classification.

Classification and Clustering Framework. To predict the popularity trajectory:

- A clustering algorithm segmented applications by meta-data similarity (unsupervised step).
- A supervised classification model (logistic regression) was trained on first-impression features to predict final popularity class.

Additionally, topic modeling (sLDA) was applied to extract dominant discussion themes from user reviews, connecting qualitative feedback with quantitative popularity outcomes.

Results. The exhaustive pipeline produced internally consistent models but limited novelty: while a statistical link between early reviews and final popularity was found, the correlation was weak and highly dependent on category biases. Over 70% of development time was spent on data structuring and feature preparation, with only marginal insight gains—an archetypal symptom of FoMO-driven Waterfall research.

Reflexive Analysis. This case exposes how the pursuit of methodological completeness (“collect everything before deciding anything”) leads to diminishing returns. The project was executed under the assumption that validation would emerge naturally from scale and precision, rather than iteration and testing. Had an Agile approach been applied—testing early hypotheses on small data subsets and refining dynamically—both insight depth and resource efficiency would likely have increased.

2.5 Agile Methodologies as Antidote to FoMO

2.5.1 Iteration Reduces Uncertainty

Agile methodologies offer frequent feedback loops that provide “mini-closures.” This reduces the cognitive anxiety associated with missing information [zhang2024fomo]. Each iteration provides bounded completeness, making uncertainty tolerable rather than threatening.

2.5.2 Psychological Agility: Embracing “Enough” Knowledge

A key psychological shift occurs when researchers redefine “complete” as “sufficient for the next iteration.” Mindfulness and self-regulation studies [zhang2024fomo] show that self-controlled decision-making mitigates FoMO by anchoring attention to present goals instead of hypothetical information gaps.

2.5.3 Team Collaboration and Shared Uncertainty

Agile frameworks emphasize communication and shared responsibility. Collaborative environments distribute FoMO—each member monitors a different informational slice. This reduces individual cognitive exhaustion and promotes continuous, collective learning, as seen in DevOps teams [4].

2.6 Integrating Research and Development Mindsets

2.6.1 From DevOps to ResOps: Research Operations

Just as DevOps transformed IT through continuous integration and delivery, a similar “ResOps” (Research Operations) approach can structure academic work. Iterative research cycles, automated feedback, and staged deliverables can reduce the psychological burden of missing out by maintaining a sense of steady progress [35].

2.6.2 Institutional Design Against FoMO

Academic institutions can adopt iterative evaluation frameworks—progress reviews, sprint-like milestones, and feedback sessions—to prevent overextension in any single phase. Such practices operationalize cognitive agility at the organizational level [11].

2.7 Conclusion: From Method to Mindset

Methodological evolution mirrors cognitive evolution. Moving from Waterfall to Agile is not merely adopting new project management tools; it is cultivating a new psychological relationship with uncertainty. By reframing iteration as progress rather than loss, researchers can overcome FoMO, regain focus, and sustain innovation where information never stops expanding.

CHAPTER 3

Pragmatic vs Cost-Efficient Data Extraction in Research: Balancing Completeness, Agility, and Risk

3.1 The Waterfall Trap in Data Extraction

3.1.1 Introduction: When “Thorough” Becomes “Pointless”

In most data-driven research projects, extraction begins with an understandable instinct: gather as much data as possible. This philosophy often manifests as a **waterfall pipeline** — where teams design a complete architecture (scraping, parsing, cleaning, storage, analysis) before ever questioning whether the collected data serves the research goal.

While such thoroughness appears methodical, it frequently produces the opposite effect. Researchers end up with massive datasets, elaborate ETL frameworks, and dashboards — yet no real insight. The process becomes an exercise in *engineering without epistemic value*.

3.1.2 The Structural Problem: A Linear Framework Detached from Meaning

A waterfall data pipeline assumes that every stage can be planned in advance, from data source to analysis. However, in research contexts — especially when the goal is exploratory — we cannot know *a priori* what is useful. The consequence is a framework that looks rigorous but collects noise.

Stage	Typical Goal	Reality in Research Context
Scraping	Capture "all possible" data (e.g., every app on Google Play)	80% of variables are irrelevant or redundant.
Cleaning	Normalize all attributes	Effort spent standardizing meaningless fields.
EDA	Search for correlations	No interpretable signal because the variables were never aligned with the research question.
Warehousing	Centralize large datasets	Adds infrastructure burden without added insight.
Reporting	Visualize metrics	Visualizations of noise still look "scientific".

3.1.3 Illustrative Use Case: The Scrapy Spider and the Hidden REST Layer

In the practical implementation, all crawling was conducted through a **Scrapy** spider that interacted indirectly with REST endpoints exposed by the target platform. Although no explicit API client was written, Scrapy’s asynchronous request manager reproduced the same network constraints as any REST-based system. This made the extraction process appear sophisticated but ultimately revealed the same pitfalls as a classical waterfall pipeline.

Gate Limits and Cascading Batches. The spider operated under strict **rate limits** imposed by the platform's servers. As a result, requests had to be queued and processed in controlled batches, each depending on the completion of the previous one. This turned the pipeline into a fragile chain of asynchronous tasks; where any delay, timeout, or pagination error propagated to the entire batch. The system was not failing technically; it was failing *conceptually*, because it treated volume as a proxy for insight.

Framework Overhead. To stabilize the process, several mechanisms were added: throttling managers, retry queues, item pipelines, and temporary caches. These additions solved short-term throughput issues but multiplied complexity. More engineering effort went into maintaining the spider's orchestration than into assessing the relevance of the collected variables. The crawler became a *well-tuned machine for gathering noise*.



The Cost of Scale and Unstructured Data. The inefficiency of the waterfall method becomes even more pronounced when dealing with **unstructured or large-scale data**. Unlike tabular datasets, information extracted from web sources, APIs, or dynamic pages lacks fixed schema and consistency. Each scraping cycle may return slightly different structures, attributes, or even semantic meanings. To handle such variability, engineers often resort to complex data warehousing architectures — introducing pipelines for cleaning, transformation, and indexing — before analysis can even begin.

However, these infrastructures come with a substantial cost. Dynamic scraping involves repeated HTTP requests, proxy rotation, pagination management, and continuous adaptation to layout changes, each of which generates redundant data and high maintenance overhead. When this raw, unstable data is warehoused prematurely, the system accumulates terabytes of noise, inflating both storage and computational expenses.

In this context, data warehousing without prior validation is not an optimization but a liability. The cost of scaling a flawed dataset grows non-linearly: the larger the volume of unverified data, the greater the waste in cleaning, storage, and energy consumption. This underlines the necessity of performing **value-oriented validation** before warehousing — transforming data storage from a default operation into a selective, knowledge-driven process.

Technical Note: Dynamic Scraping and Its Operational Friction. Large-scale data extraction from online platforms often relies on **scraping frameworks** such as *Scrapy*, which manage asynchronous HTTP requests, response parsing, and data pipelines. In practice, these systems must navigate technical constraints imposed by servers to avoid rate limits or IP bans. This is typically achieved through the use of **proxies** — intermediary servers that relay requests — and **proxy rotation**, which distributes requests across multiple IP addresses to mimic distributed traffic.

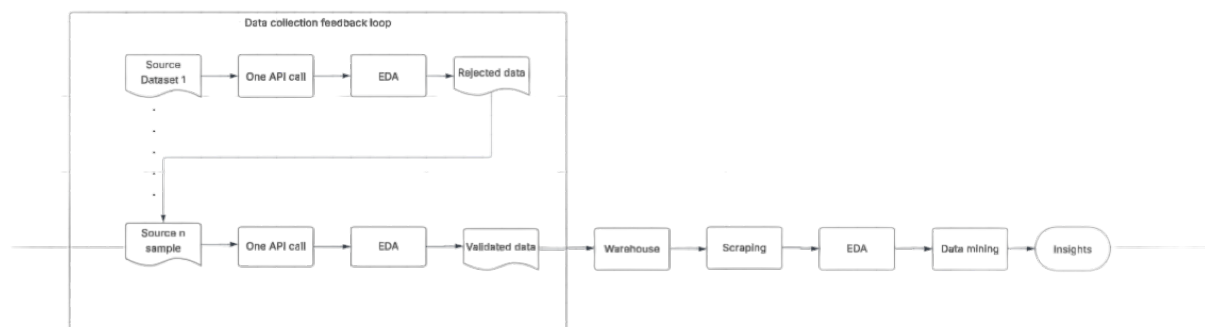
While these mechanisms increase throughput and reduce blocking risk, they also add substantial engineering overhead. Each layer (proxy pool management, retry policies, and asynchronous queuing) introduces potential failure points and raises the operational cost of maintaining the crawler. Scrapy’s modular architecture amplifies this complexity: spiders, middleware, item pipelines, and scheduling queues must all be tuned to keep pace with changing site structures and data volumes. In a research context, these optimizations improve technical robustness but rarely improve epistemic value — they make the system more resilient at collecting *irrelevant* data. The more sophisticated the scraping infrastructure becomes, the further it drifts from its initial purpose: collecting information that actually answers the research question.

Key Insight. The Scrapy case demonstrates that technical refinement cannot compensate for poor epistemic design. Even when the framework manages REST calls efficiently, the absence of an early feedback loop about data relevance leads to a full-scale system that works perfectly — on data that does not matter.

3.1.4 The Full Waterfall Knowledge Discovery Process

The waterfall logic extended far beyond data extraction. After storage, the project continued through the entire **Knowledge Discovery from Data (KDD)** chain — including exploratory analysis, clustering, and descriptive modeling — before realizing that the underlying data itself lacked value.

3.1.5 The Pragmatic Alternative: Value-First Iterative Process



The improved model introduces an **early feedback loop** between the research question, data selection, and acquisition steps. Instead of executing the entire KDD pipeline blindly, the process iteratively validates whether each batch of extracted data contributes meaningfully before proceeding to analysis.

3.1.6 The Value of Post-Validation Warehousing and Processing.

In the value-first approach, data warehousing and subsequent processing stages occur *only after* the validation of data relevance and quality. This inversion of order is critical: in the traditional waterfall framework, raw and unverified data were prematurely stored and transformed, multiplying redundancy and noise. In contrast, the validated data entering the warehouse already carries semantic alignment with the research objective, making every downstream operation — aggregation, indexing, and feature engineering — inherently more efficient and interpretable.

By deferring heavy computation and storage until validation, the process eliminates the cost of maintaining useless records, reduces schema drift, and ensures that processing power is applied to information with proven analytical potential. Thus, the warehouse ceases to be a passive data dump and instead becomes an **active layer of knowledge curation**, hosting only datasets that have passed empirical and contextual checks. The outcome is a pipeline where computational efficiency and epistemic value converge: the system processes less data, but produces more meaningful insights.

3.2 Guides for Evaluating Data Efficiency and Quality

Assessing whether data is “good enough” for analysis requires both practical heuristics and theoretical grounding. This section provides two complementary views: a pragmatic guide for day-to-day research practice, and an extensive framework based on established data-quality literature.

3.2.1 1. Practical Guide: Minimal Evaluation for Research Pipelines

In most empirical or data-driven studies, researchers cannot afford exhaustive data-quality audits. Instead, they rely on lightweight, repeatable checks that detect structural or semantic issues early — a concept consistent with the “fitness-for-use” paradigm proposed by Wang and Strong [50]. The focus is on preventing wasted computation and ensuring the dataset is coherent enough to yield reproducible findings.

Key Heuristics.

- **Noise vs. Signal:** Evaluate whether patterns persist after basic denoising or aggregation. If correlations disappear, the dataset may be too noisy to justify modeling.
- **Coverage:** Verify that the data span the relevant entities, languages, or time windows, and that no single source dominates more than roughly 70% of the sample.
- **Bias Checks:** Look for obvious cultural, selection, or platform bias. Document such biases following the “Datasheets for Datasets” framework [12].
- **Replicability:** Ensure another researcher could reproduce the extraction and preprocessing steps with minimal divergence [Schelter2018].
- **Semantic Alignment:** Confirm that the variables or fields directly relate to the hypothesis or target concept, not merely to source availability.
- **Stability Over Time:** For streaming or scraped data, compare new batches to prior ones for drift in key distributions.

Quick Sanity Techniques.

- Run exploratory plots and summary statistics to confirm expected distributions.
- Inspect random samples for malformed entries or missing values (should be below 5–10% overall).
- Fit a trivial baseline model; if it performs no better than random, re-evaluate data relevance.
- Manually review a small batch of outputs for contextual meaning and labeling errors.

Outcome. These heuristics yield a binary decision: either *the data are stable and meaningful enough to proceed*, or they are not. This lightweight evaluation provides a stopping rule before the costly steps of warehousing and model training, thereby improving what Pipino, Lee, and Wang [38] describe as “pragmatic data-quality assessment.”

3.2.2 2. Extensive Guide: Conceptual and Metric Framework

For documentation, audits, or publication, a more formal evaluation may be required. This second guide synthesizes standardized quality dimensions, measurable indicators, and process-oriented evaluation methods from information-quality literature [2, 20, 22].

Core Dimensions (after ISO/IEC 25012 and Batini & Scannapieco).

- **Accuracy:** recorded values match the real-world entities or events they represent.
- **Completeness:** all required fields and records exist; coverage of target population is adequate.
- **Consistency:** no conflicts among datasets or across temporal updates.
- **Timeliness:** data are current relative to the phenomena under study.

- **Validity:** values conform to domain rules, formats, and logical constraints.
- **Uniqueness:** duplicates are eliminated and identifiers remain stable.
- **Integrity:** relationships between entities are logically correct and maintained.
- **Relevance:** variables are meaningfully connected to the research objective.

Quantitative Metrics (adapted from Pipino, Lee, and Wang).

$$\text{Completeness} = \frac{\#\text{non-null}(A)}{\#\text{records}} \quad (3.1)$$

$$\text{Validity} = \frac{\#\text{values satisfying constraints}}{\#\text{records}} \quad (3.2)$$

$$\text{Duplication Rate} = \frac{\#\text{duplicates}}{\#\text{records}} \quad (3.3)$$

$$\text{Consistency Score} = 1 - \frac{\#\text{violations}}{\#\text{checks}} \quad (3.4)$$

Such metrics can be computed automatically through profiling scripts or data-validation libraries. A comprehensive evaluation would further include a relevance proxy (e.g., proportion of variance explained by key variables) to ensure the data carry empirical signal.

Process View. Kahn, Strong, and Wang [22] propose treating information quality as a continuous product-and-service process: data are collected, validated, consumed, and refined iteratively. From this perspective, efficiency means minimizing the cost of correction while maximizing interpretability across reuse contexts.

Synthesis. The extensive guide provides a defensible vocabulary for formal reporting, while the practical guide in Section 3.2.1 supports iterative, resource-aware validation during collection. Together they satisfy both the methodological rigor expected in publications and the agility demanded by real-world research pipelines.

3.2.3 Conclusion

The waterfall approach to data extraction seduces researchers with its promise of completeness and structure, yet it often produces frameworks that operate flawlessly on data that **does not matter**. A pragmatic, value-oriented iteration process acknowledges that usefulness, not quantity, is the true measure of data quality. Rather than optimizing throughput, we optimize **meaning**; saving time, resources, and analytical integrity.

CHAPTER 4

Purpose-Driven Evaluation and Benchmarking in Iterative Research

4.1 Introduction

Evaluation is traditionally presented as a neutral, objective step in the research process. Yet work in evaluation theory, design science research, cognitive psychology, and empirical software engineering all converge on a different picture: evaluation is *contextual*, *purpose-driven*, and never fully neutral [13, 23, 35, 39]. In parallel, DevOps practices have shown that assessment must evolve with the system under development rather than remaining fixed [35].

This chapter pursues two objectives:

1. use existing scientific literature to expose the limitations of *Waterfall-style* evaluation (fixed metrics, static baselines, one-shot validation);
2. propose a **Purpose-Oriented Benchmark Framework** and instantiate it in the “First Impression” use case, combining formal definitions with a lightweight, empirically realistic implementation.

4.2 The Myth of the Universal Metric

Evaluation theory (Scriven, Patton) and design science research argue that no evaluation metric is universally correct; it must reflect the purpose and context of the study. Cognitive and behavioural research further shows that the search for a single “right” metric is often a manifestation of **Fear of Missing Out (FoMO)** and perfectionism, associated with procrastination, reduced self-control, and inefficient decision-making [27, 42].

In practice, the belief in a universal metric leads to:

- over-optimisation on a narrow performance measure,
- confirmation bias toward initial hypotheses,
- a false sense of significance,
- rigid evaluation pipelines that resist iteration or reframing.

The literature therefore, supports a central claim for this chapter: **metrics must be purpose-driven, not template-driven.**

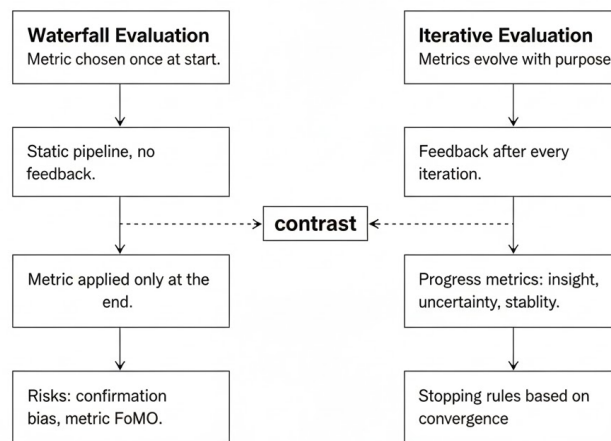
4.3 The Problem With Waterfall Evaluation

In the Waterfall paradigm, evaluation is:

1. defined once, at the beginning of the project;
2. applied once, at the end;
3. disconnected from insights that emerge during the study.

Kahneman and Tversky describe how commitment to an early framing produces **confirmation bias** and sensitivity to sunk costs [23]. This also interacts with FoMO: researchers fear choosing the “wrong” metric and missing the “true” result, which encourages metric-shopping and complex, one-shot evaluation designs.

Scriven characterises this as a “terminal metric trap”: evaluation degenerates into mere outcome scoring. Design science research and DevOps both show that this is incompatible with systems that evolve through cycles of construction, feedback, and refinement.



4.4 Why Iterative Research Needs Iterative Evaluation

Agile research practices, DevOps continuous assessment, and developmental evaluation models converge on a shared principle:

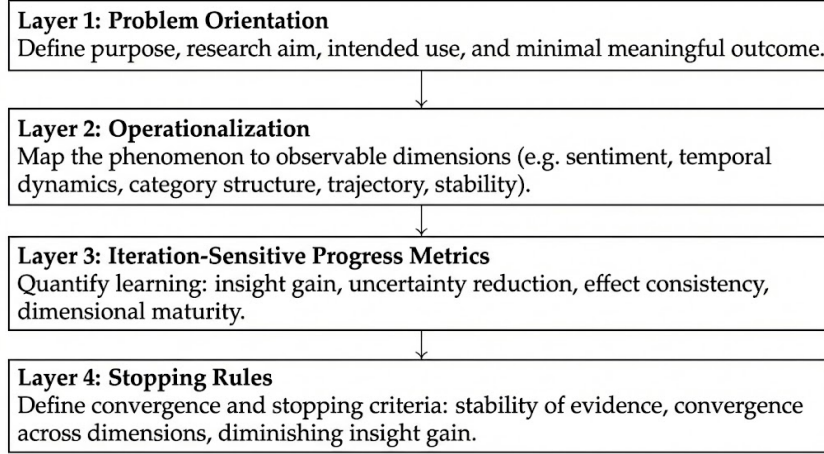
Evaluation must change as the system and understanding change.

Rather than providing a single final verdict, evaluation in an iterative framework:

- Tracks the evolution of understanding,
- Detects convergence or divergence of effects,
- Indicates when a method should be refined, pivoted, or stopped,
- Acts as a feedback loop, not a final exam.

To structure this, we propose a four-layer **Purpose-Oriented Benchmark Framework**.

4.4.1 Purpose-Oriented Benchmark Framework



4.5 Formal Benchmark Dimensions and Measures

For high-sensitivity contexts (e.g. when optimising models where even a 2% improvement matters), the four benchmark dimensions of Layer 3 can be defined formally.

Assume the research process proceeds in iterations $t = 1, 2, \dots, T$.

4.5.1 Insight Gain

Let M_t be a model or analytical specification at iteration t , and R_t^2 its explanatory power (or any suitable global fit index).

Definition. The *insight gain* between iterations $t - 1$ and t may be approximated as:

$$IG_t = R_t^2 - R_{t-1}^2. \quad (4.1)$$

Alternatively, if $\mathcal{F}_t$ is the set of features deemed significant at iteration t :

$$IG_t = |\mathcal{F}_t \setminus \mathcal{F}_{t-1}|. \quad (4.2)$$

4.5.2 Uncertainty Reduction

Let $\hat{\theta}_t$ be an estimated effect (e.g. the coefficient for the first-impression predictor) and $\text{Var}_{\text{boot}}(\hat{\theta}_t)$ its bootstrap variance.

Definition. Uncertainty reduction between iterations can be written as:

$$UR_t = \text{Var}_{\text{boot}}(\hat{\theta}_{t-1}) - \text{Var}_{\text{boot}}(\hat{\theta}_t), \quad (4.3)$$

or, using confidence intervals CI_t :

$$UR_t = \text{width}(CI_{t-1}) - \text{width}(CI_t). \quad (4.4)$$

4.5.3 Effect Consistency

Let $\beta_t^{(k)}$ denote the estimated effect in condition or subset k at iteration t (e.g. per category or cohort).

Definition. A simple coefficient stability index is:

$$CSI_t = 1 - \frac{|\beta_t - \beta_{t-1}|}{|\beta_{t-1}| + \varepsilon}, \quad (4.5)$$

with ε a small constant for numerical stability. A replicability score across K resampled datasets is:

$$Rep_t = \frac{1}{K} \sum_{k=1}^K \mathbf{1}(\text{sign}(\beta_t^{(k)}) = \text{sign}(\beta_t)), \quad (4.6)$$

where $\mathbf{1}(\cdot)$ is the indicator function.

4.5.4 Stability Under Perturbation

Let m_i be a performance measure (e.g. R^2 , AUC, or RMSE) computed on a perturbed dataset D_i (e.g. via resampling or cohort-based cross-validation).

Definition. A simple stability index is:

$$SI = 1 - \frac{\text{Var}(m_i)}{\text{Var}(m_i) + \varepsilon}, \quad (4.7)$$

with $SI \in [0, 1]$ (higher indicates more stability).

4.5.5 Stopping Rules in Formal Settings

In high-sensitivity optimisation, stopping may be defined by thresholds such as:

$$|IG_t| < \epsilon_1, \quad (4.8)$$

$$UR_t < \epsilon_2, \quad (4.9)$$

$$Rep_t > 1 - \epsilon_3, \quad (4.10)$$

$$SI > 1 - \epsilon_4, \quad (4.11)$$

for small tolerances $\epsilon_i > 0$.

Applicability to Empirical Studies

These formal measures are appropriate when:

- small numerical differences have operational impact (e.g. model selection in production systems),
- statistical optimisation is the primary goal.

However, in an empirical exploratory study such as ours, applying the full suite at every iteration would:

- recreate Waterfall-like behaviour,
- inflate the cost of iteration,
- reinforce metric-oriented FoMO.

For this reason, the practical implementation in the “First Impression” project adopts a **lightweight operationalization**, which preserves the spirit of the framework without turning evaluation into a separate optimisation problem.

4.6 Lightweight Operationalization in Practice

In the concrete workflow, each iteration (e.g. change in clustering strategy or model structure) is evaluated using four pragmatic axes:

1. **Relevance:** does the structure obtained reveal meaningful differences in the dependent variables (long-term popularity, trajectories, etc.)?
2. **Interpretability:** can the result be expressed in clear, domain-relevant terms (e.g. “high early sentiment and frequent updates” vs. “low sentiment and rare updates”)?

3. **Consistency:** do the patterns persist under simple resampling or restriction to subsets (e.g. specific categories)?

4. **Cost–benefit:** is the conceptual insight gained proportionate to the analytical effort invested?

Conceptually, these axes correspond to the formal dimensions of insight gain, uncertainty reduction, and stability, but at a resolution compatible with an Agile, empirical research workflow.

4.7 Use Case: Clustering by Dependent Variables

A key analytical task in the “First Impression” study was to cluster applications according to their dependent variables (long-term popularity metrics, rating trajectories, update cadence). The objective was to identify patterns of behaviour and test whether early sentiment helped to explain membership in these groups.

Use Case: Clustering Applications by Dependent Variables

Iteration 1: Unsupervised Clustering (k-means)

The first attempt used k-means clustering on a feature space built from dependent variables (e.g. average rating, rating variance, trajectory features).

Evaluation under the lightweight benchmark:

- **Relevance:** low. Let μ_c denote the mean long-term popularity metric in cluster c . The between-cluster differences $\mu_c - \mu_{c'}$ were small and often non-significant, indicating weak separation in terms of the outcome of interest.
- **Interpretability:** low. Cluster centroids did not map to interpretable behavioural profiles (e.g. no clear “high-stable” vs. “low-volatile” pattern).
- **Consistency:** weak. Running k-means on category-specific subsets yielded different numbers and types of clusters, and the direction of any observed difference in popularity was unstable.
- **Cost–benefit:** poor. Considerable time was spent choosing k , scaling schemes, and distance metrics, with limited conceptual return.

Within the Purpose-Oriented Benchmark Framework, this iteration is recorded as a *useful negative result*: the numerical structure induced by k-means on the dependent variables is not aligned with the conceptual structure of “first-impression influence”.

Iteration 2: Decision-Rule Clustering

The second attempt replaced k-means with a decision-rule clustering constructed from domain logic. For example:

Group A : high early sentiment $\wedge$ frequent updates,

Group B : low early sentiment $\wedge$ rare updates,

Group C : volatile early sentiment $\wedge$ irregular updates.

Evaluation under the same axes:

- **Relevance:** high. Let $\mu^{(A)}$, $\mu^{(B)}$ denote mean long-term popularity in Groups A and B. We observe $\mu^{(A)} > \mu^{(B)}$ with consistent effect sizes across samples, directly supporting the research question.
- **Interpretability:** very high. Each group is expressed in domain terms, linking first-impression signals and maintenance patterns to outcome trajectories.
- **Consistency:** strong. When the analysis is restricted to specific categories, the direction of effects (e.g. Group A outperforming Group B) remains stable.
- **Cost–benefit:** excellent. The effort required to design and implement the decision rules produced a substantial increase in conceptual clarity and explanatory power.

Outcome

This comparison illustrates the role of the benchmark: the goal is not to crown a universally “best” clustering algorithm, but to identify which iteration *advances understanding* of the phenomenon. Under the Purpose-Oriented Benchmark Framework, the decision-rule clustering clearly dominates k-means in terms of relevance, interpretability, consistency, and cost–benefit, and is therefore retained for subsequent analyses.

4.8 Benchmark Specification: First-Impression Predictiveness Benchmark (FIPB)

Building on the four-layer framework and the use case, we define the **First-Impression Predictiveness Benchmark (FIPB)** as a multi-dimensional benchmark tailored to the question: *Does early user sentiment provide robust predictive value for long-term application performance?*

Benchmark Dimension	Description and Rationale
Sentiment Signal	Polarity and intensity of early reviews. Identifies whether a first-impression signal exists and is measurable.
Temporal Dynamics	Interaction between early sentiment and update cadence; tests whether the timing and frequency of updates modulate the first-impression effect.
Category Structure	Controls for domain-level biases across app categories; ensures that apparent effects are not artefacts of genre or market segment.
Trajectory Evolution	Characterises volatility or stability in long-term rating and popularity trajectories; evaluates whether early sentiment predicts the shape of these trajectories.
Stability Under Resampling	Assesses robustness of findings across random subsamples, cohort reshuffling, and temporal splits; ensures that conclusions do not depend on a single partition of the data.

FIPB thus combines the conceptual layers of the framework (purpose, operationalization, progress metrics, stopping rules) with concrete dimensions grounded in the empirical context.

4.9 Conclusion

The literature demonstrates that there is no universal metric, no context-free evaluation, and no justification for treating evaluation as a fixed, post-hoc procedure. Evaluation theory, design science research, cognitive psychology, and DevOps jointly argue for **iterative, purpose-driven** evaluation.

This chapter responded to that call by:

- Articulating the limitations of Waterfall-style evaluation and the role of FoMO in reinforcing metric rigidity;
- Proposing a four-layer Purpose-Oriented Benchmark Framework;
- Providing formal definitions of progress metrics and stopping rules for high-sensitivity optimisation settings;
- Introducing a lightweight operationalization suitable for empirical, exploratory research;
- Instantiating the framework in the First-Impression Predictiveness Benchmark (FIPB) and illustrating its use through a clustering case study.

In this perspective, evaluation ceases to be a terminal verdict and becomes a **compass for iterative research**, guiding when to refine, when to pivot, and when to stop.

PART II

Empirical Use Case — First Impressions and App Popularity

CHAPTER 5

The use case: Empirical study on the impact of user's feedback on the

In this project we are aiming to demonstrate the use of different methodologies and data processing and their effect on the success and impact of research. We are using the case study as example (or counter example) on the conduct of research.

5.1 Introduction

The advent of agile methodologies has revolutionised the landscape of mobile application development. Characterised by iterative development cycles, agile methodologies have enabled developers to respond quickly to changes and deliver high-quality applications. This approach has significantly improved the efficiency of application development, leading to the creation of more robust and user-centred applications. However, the evolution of application development should not stop at the implementation of agile methodologies. There is a continuous need for optimisation and improvement to ensure that mobile applications meet the ever-changing demands of users and the dynamics of the technological landscape. To this end, various release strategies have been developed to further improve the efficiency of application development and delivery. At the heart of agile methodologies lies the principle of user interaction and feedback.

Agile development emphasises close collaboration with users and stakeholders and iterative feedback loops to continuously improve the product. In the context of mobile applications, user reviews serve as a crucial channel for obtaining feedback. These reviews provide invaluable insights into user experiences, preferences, and issues, which can inform mobile application development and release strategies.

Therefore, while agile methodologies and release strategies have significantly improved application development, there is a need to leverage user reviews more to optimise these strategies.

5.1.1 Current understanding of release strategies:

The release strategy is a key element in mobile application development. It defines the timing and frequency of application updates, which can have a significant impact on user satisfaction and the success of the application. Most successful applications start with a solid release strategy, often based on time.

In their study entitled 'Release Practices for Mobile Apps – What do Users and Developers Think?' [Nay16], Nayebi, Adams, and Ruhe explored the different release strategies used by mobile application developers. They identified six distinct release strategies, most of which are time-based.

However, a key finding of their research is that the majority of developers are willing to adjust their time-based strategies in response to user feedback. As they noted, "the majority of developers are willing to modify their time-based strategies to take user feedback into account" [Nay16]. This finding highlights suggests that, although time-based launch strategies are commonly used, they are not necessarily the best approach for all applications or contexts.

Whether user feedback can provide valuable insights that can help refine the launch strategy and improve the application is under studied. Our goal is to develop a robust launch strategy based on user feedback that can help developers improve their applications and maximise their success.

5.1.2 Factors influencing app updates:

In the context of mobile app development, understanding the factors that influence app updates is essential for optimising launch strategy. Regular updates are often perceived as a sign of developer commitment and can positively influence user perception and app popularity.

In their study "Fresh apps: an empirical study of frequently-updated mobile apps in the Google Play Store" [McI15], McIlroy, Ali, and Hassan explored the factors that influence mobile application updates. They found that frequently updated applications generally follow a timing strategy, while those that are not frequently updated are inconsistent in their approach. This suggests that implementing a consistent update strategy can be a key success factor for a mobile app.

However, their research also revealed that 25% of the changes made during updates are negative. This raises the question of whether we can predict the direction of changes based on user feedback. This question is particularly relevant to our research, as we seek to develop a launch strategy based on user feedback.

In summary, this study focusing on what the google apps store data can give us about the importance of a consistent and frequent update strategy for the success of a mobile application. It also highlights the need for a better understanding of the impact of updates on users and how user feedback can be used to inform the update strategy. Our research aims to methodologize filling this gap by exploring how user feedback can be used to develop a more effective release strategy.

5.1.3 App launch timing and success In the field of mobile app development

Optimising the launch strategy is key to maximising an app's success. In their study "Towards Release Strategy Optimisation for Apps in Google Play" [She17], Shen, Lu, Hu and Liu explored how to help developers choose the right time to launch their applications in order to maximise positive feedback from users.

They discovered that the average update frequency is not always a comprehensive indicator of an application's launch strategy. Instead, they suggest dividing the update interval into three groups: successive, normal and rare, and finding differences in user feedback.

Their analysis reveals that the change in an app's ranking after a successive update suggests the Matthew Effect. They state: "Analysis results reveal that the change of app ranking after a successive update suggests the Matthew Effect, i.e., the high-ranked apps get higher ratings and the low-ranked apps get lower ratings."

This suggests that the timing of an update can have a significant impact on the success of the app. Apps that are already highly ranked may benefit more from an update, while those that are lower ranked may see their ranking drop even further. This highlights the importance of choosing the right time to release an update. But the interaction between the two variables, time and user_feedback is still to be explored.

5.1.4 Gaps in the literature and psychology of first impressions:

The primary objectives of the use case are to:

- Evaluate how early user input influences the deployment plans of mobile apps, allowing developers to modify their upgrade tactics to enhance engagement and app uptake.
- Create a robust plan for releasing your app that takes advantage of user input to improve performance and increase popularity.

The aim of the use case is an establishment of a beneficial loop in which user input continuously guides future app enhancements, ultimately boosting user commitment and competitiveness in the industry.

- The challenge we are demonstrating is in the use of web data to conduct empirical research, replacing the need for surveys. However, we suggest that the human response in surveys are highly meaningful, human behavioral psychology evolve with time and continuously studying it is irreplaceable. The idea is using the web as another mean, an evaluation of online behaviors and then linking them to real life psychology.

However, this lack in literature is not a result of lack of interest of researchers. We believe that to get meaningful data to explain the psychology of the user, a thorough behavioral and psychological surveys

throughout at least three updates; and conducting such research on a significant sample of applications from different level of initial success and potential is time and cost expensive.

The focus of this study changed from getting knowledge about the user first impression to studying the data and processing for empirical research from web data using techniques like PCA, kmeans, sLDA etc.. after noticing the barriers affecting this research.

5.2 Problem Formalization

This section formalizes the empirical problem addressed in the study. The objective is to characterise and analyse the relationship between early user feedback on mobile applications and their long-term popularity using large-scale observational data.

5.2.1 Study Entities and Notation

Let $\mathcal{A} = \{a_1, a_2, \dots, a_N\}$ denote the set of mobile applications considered in the study. For each application a_i , we observe a sequence of user interactions and application metadata over time, derived from the Google Play Store.

Time is represented discretely. Let $t = 0$ denote the release time of an application, and let $t = 1, 2, \dots, T_i$ index subsequent observation windows for application a_i .

For each application a_i and time window t , we observe:

- a set of user reviews $R_i(t) = \{r_{i,1}(t), \dots, r_{i,n_i(t)}(t)\}$,
- a corresponding set of ratings $S_i(t) = \{s_{i,1}(t), \dots, s_{i,n_i(t)}(t)\}$, where $s_{i,j}(t) \in \{1, 2, 3, 4, 5\}$,
- application metadata including update events and category labels.

5.2.2 Dependent Variable: Application Popularity

The dependent variable of interest is the long-term popularity of an application. Popularity is not directly observable as a single quantity, but is approximated using aggregate signals derived from user interaction over time.

Let P_i denote a scalar popularity indicator for application a_i , computed over a sufficiently long time horizon $[t_L, T_i]$ following initial release. In this study, P_i is operationalised using one or more of the following aggregated measures:

- the mean rating over the long-term window,
- the cumulative number of ratings,
- the stability or trend of the rating trajectory.

These indicators are treated as proxies for user adoption and sustained engagement, consistent with prior empirical work on mobile application success.

5.2.3 Independent Variables: Early User Feedback

The primary independent variables capture early user feedback following application release. Let τ denote a fixed early observation window (e.g. the first days or weeks after release).

For each application a_i , early feedback is represented by:

- the set of early reviews $R_i^{(0)} = R_i(t)$ for $t \leq \tau$,
- the corresponding early ratings $S_i^{(0)} = S_i(t)$ for $t \leq \tau$.

From these data, a feature vector $X_i^{(0)} \in \mathbb{R}^d$ is constructed, encoding early user sentiment and interaction patterns. Components of $X_i^{(0)}$ include, but are not limited to:

- summary statistics of early ratings (mean, variance),
- temporal characteristics of early rating accumulation,
- latent features extracted from review text.

5.2.4 Operational Definition of First Impression

In the context of this study, the notion of *first impression* is operationalised as the information available to users and developers within the early observation window $[0, \tau]$. Formally, the first impression of application a_i is represented by the feature vector $X_i^{(0)}$.

This definition reflects the earliest observable feedback signals following release and is aligned with the way users encounter and assess applications in digital marketplaces.

5.2.5 Problem Statement

The empirical problem addressed in this study can be stated as follows.

Given:

- early feedback features $X_i^{(0)}$ for each application a_i ,
- long-term popularity indicators P_i ,

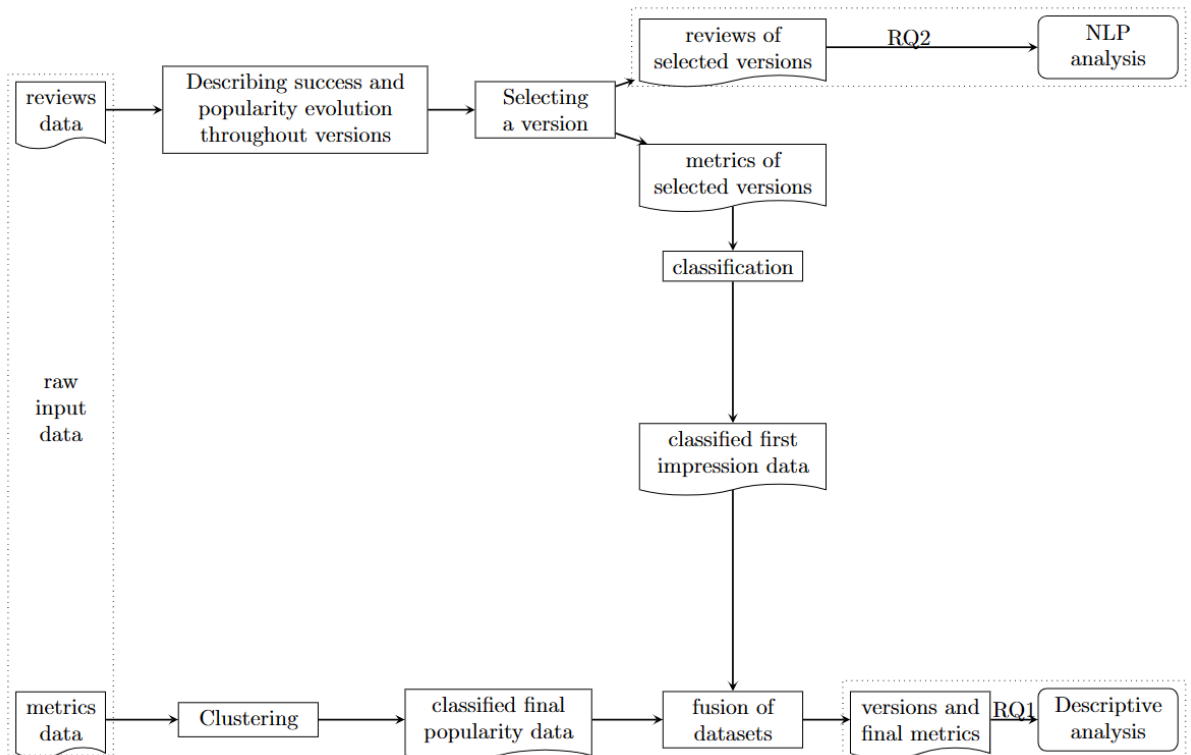
we seek to investigate whether structure, variation, or patterns in $X_i^{(0)}$ are associated with differences in P_i .

Rather than assuming a specific functional form *a priori*, the analysis explores this relationship through dimensionality reduction, clustering, and probabilistic modeling techniques. These methods aim to identify latent structure in early feedback data and assess how such structure relates to long-term application outcomes.

5.3 Research Questions and Analytical Scope

The empirical analysis conducted in this study is guided by a set of explicit research questions that structure the data collection, processing, and analytical workflow. These questions define the scope of the investigation and determine how observational data from mobile application stores is transformed into measurable quantities and analytical outputs.

The analytical process follows a structured pipeline, illustrated in Figure 5.1, which links raw data acquisition to the identification of factors associated with application success and popularity. Each stage of this pipeline corresponds to one or more research questions, ensuring that methodological steps are directly tied to empirical objectives.



5.3.1 Research Questions

The study addresses the following research questions:

- **RQ1:** Do early user feedback signals associated with mobile applications exhibit structured patterns that can be extracted from large-scale app store data?
- **RQ2:** Are patterns observed in early user feedback associated with differences in long-term application popularity?
- **RQ3:** Which observable factors derived from reviews, ratings, and update histories contribute most to distinguishing application popularity profiles?

These questions are exploratory in nature. They do not assume a specific functional relationship between early feedback and long-term outcomes, but rather seek to identify structure, variation, and associations present in the data.

5.3.2 Analytical Scope

The scope of the analysis is defined by observational data collected from the Google Play Store. The study focuses on applications for which sufficient user feedback is available to characterise early user responses, as well as subsequent application evolution. To this end, applications are filtered based on criteria such as the availability of multiple versions and a minimum number of user reviews, ensuring that both early and longer-term signals can be observed.

Early user feedback is operationalised through ratings and textual reviews collected shortly after application release. Long-term popularity is approximated using aggregated indicators derived from rating trajectories and user interaction patterns observed over an extended period. The analysis is restricted to information that is publicly available, and no survey-based or self-reported data is used.

5.3.3 Mapping Research Questions to the Analytical Pipeline

The relationship between the research questions and the analytical workflow is summarised as follows. Data collection and preprocessing stages provide the raw material required to address **RQ1** by enabling the extraction of early feedback features. Dimensionality reduction and clustering techniques are applied to uncover latent structure within these features, directly supporting **RQ1** and **RQ2**. Finally, factor identification and comparative analysis are used to examine how extracted features relate to long-term popularity indicators, addressing **RQ3**.

By structuring the analysis around these research questions, the study ensures that each methodological step contributes directly to answering a well-defined empirical objective. The subsequent sections of this chapter provide a detailed technical description of the data representation, feature construction, and analytical methods employed within this framework.

5.4 Data Representation and Feature Engineering

The data consists of heterogeneous signals that must be converted into quantitative features before analysis. Feature engineering focuses on constructing indicators of early user feedback and long-term popularity while preserving temporal consistency.

5.4.1 Version Selection Strategy

Applications evolve through multiple versions, and feedback is unevenly distributed across release histories. Selecting versions based on score or review volume introduces bias, as these metrics are themselves outcomes of popularity.

To avoid this bias, versions are selected using time-framed criteria. Only versions located within a defined temporal window relative to the application release history and exceeding a minimum review threshold are considered. This ensures statistical viability while preserving early exposure conditions.

5.4.2 Rating-Based Feature Construction

Ratings provide numerical indicators of user satisfaction and are directly available from the platform. For each retained application, ratings submitted within the early observation window are aggregated to construct rating-based features.

The following features are computed:

- mean early rating,
- variance of early ratings,
- total number of early ratings,
- rate of rating accumulation within the early window.

These quantities are defined for all retained applications and exhibit sufficient variability across application categories and popularity levels. Mean ratings span comparable ranges across categories, while variance captures differences in early user consensus.

5.4.3 Textual Review Representation

User reviews provide unstructured textual feedback reflecting user experience and perception. Review text submitted within the early observation window is preprocessed using standard natural language processing steps, including tokenization, stop-word removal, and text normalization.

For each application, early reviews are aggregated and represented using a document-term structure. At this stage, no unexpected or directly interpretable insights are observed from the textual content alone. The representation is therefore retained in a preparatory form, intended for potential use in subsequent topic modeling analyses such as Latent Dirichlet Allocation (LDA), conditional on the existence of stable application groupings.

5.4.4 Integrated Analytical pipeline:

Given the data and feature construction choices described above, we designed an integrated analytical pipeline combining unsupervised and supervised components.

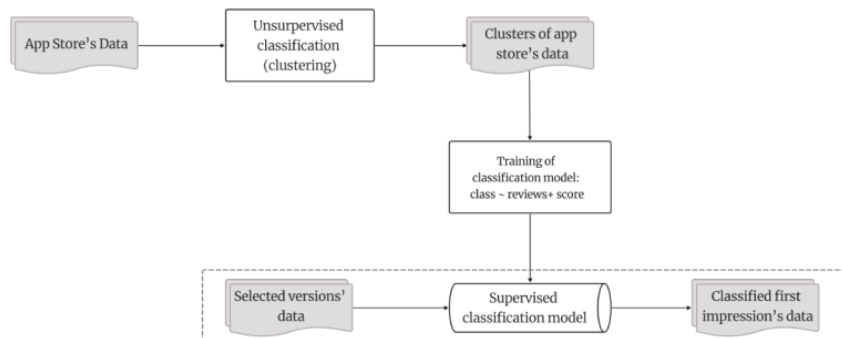


Figure 5.2 summarizes the conceptual analytical pipeline guiding this study. While this design motivated the initial methodological choices, each component was subsequently evaluated on empirical and diagnostic grounds rather than assumed to be valid a priori.

This system-level design represents the intended integration of clustering-based structure discovery with classification-based prediction, whose individual components are evaluated and diagnostically assessed in subsequent sections.

5.5 Dimensionality Reduction and Latent Structure (Diagnostic Analysis)

This section investigates the structural properties of the engineered feature space described in Section 4. Rather than aiming at discovering latent behavioral patterns or optimizing predictive performance, the objective is to assess variance concentration, feature redundancy, and the feasibility of unsupervised structural modeling given the current feature construction.

Dimensionality reduction and clustering methods are therefore employed strictly as *diagnostic tools*, used to evaluate whether the feature space admits meaningful low-dimensional representations or stable partitions before committing to further modeling stages.

5.5.1 Purpose and Scope of Diagnostic Analysis

High-dimensional feature spaces constructed from web-derived metrics are particularly susceptible to redundancy, scale dominance, and spurious structure induced by correlated variables. In such contexts, applying unsupervised methods without prior diagnostic evaluation risks producing misleading interpretations and overfitted analytical narratives.

The analyses presented in this section are designed to answer the following methodological questions:

- Is variance concentrated along a small number of dominant directions?
- Do the engineered features encode separable structural patterns or continuous gradients?
- Are unsupervised partitioning methods appropriate given the observed data geometry?

Answering these questions serves to inform subsequent methodological decisions, including whether to refine feature engineering, revise grouping strategies, or halt further unsupervised exploration.

5.5.2 Principal Component Analysis as a Diagnostic Tool

Mathematical Formulation

Principal Component Analysis (PCA) is applied to the standardized feature matrix $X \in \mathbb{R}^{N \times d}$, where each feature is centered and scaled to unit variance to mitigate scale effects.

Let Σ denote the empirical covariance matrix:

$$\Sigma = \frac{1}{N-1} X^\top X.$$

PCA consists of solving the eigenvalue problem:

$$\Sigma \mathbf{u}_k = \lambda_k \mathbf{u}_k,$$

where λ_k represents the variance explained by the k -th principal component and $\mathbf{u}_k$ its associated eigenvector. The data can then be expressed in the principal component basis via the projection:

$$Z = XU,$$

where $U = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_d]$.

In this study, PCA is not employed to construct a reduced embedding for interpretation or downstream modeling, but to diagnose the variance structure induced by the engineered features.

Variance Distribution and Redundancy Assessment

The eigenvalue spectrum obtained from PCA provides insight into the intrinsic dimensionality of the feature space. The corresponding scree plot and cumulative explained variance indicate that a limited number of components capture a substantial portion of the total variance, while no single component dominates the representation.

This gradual decay of eigenvalues reflects the presence of strong correlations among features, particularly those related to engagement scale and popularity magnitude. The inclusion of interaction terms further amplifies this effect by reinforcing existing variance patterns rather than introducing orthogonal sources of information.

These observations suggest that the feature space is characterized by redundant dimensions and continuous variation rather than compact, low-dimensional structure.

Limits of Interpretation and Decision Not to Project

Although PCA is commonly accompanied by low-dimensional visual projections, such projections are only meaningful when the underlying feature space admits interpretable orthogonal structure. In the present case, diagnostic analysis reveals that the dominant principal components are primarily driven by correlated scale-related variables and their interactions.

Projecting the data onto the first principal components would therefore reflect magnitude effects and feature construction artefacts rather than intrinsic behavioral structure. Such visualizations risk creating an illusory sense of separation or interpretability unsupported by the data.

For this reason, PCA is retained solely as a diagnostic instrument for variance and redundancy assessment, and low-dimensional projections are deliberately omitted. This decision prioritizes methodological transparency and avoids over-interpretation of structurally uninformative representations.

5.5.3 Clustering as a Structural Stress Test

k-means Assumptions and Application

As a complementary diagnostic approach, k-means clustering is applied to the standardized feature space to test the hypothesis that applications form naturally separable groups based on early user feedback.

The k-means algorithm partitions the data into k clusters by minimizing the within-cluster sum of squared distances:

$$\sum_{i=1}^N \|x_i - \mu_{c(i)}\|^2,$$

where $\mu_{c(i)}$ denotes the centroid of the cluster assigned to observation i . Implicit in this formulation are assumptions of compact, spherical clusters with comparable variance.

Clustering is evaluated across multiple values of k to assess robustness and sensitivity to hyperparameter choice.

Sensitivity to k and Lack of Stable Partitioning

The clustering outcomes exhibit strong sensitivity to the choice of k , with cluster assignments varying substantially across configurations. Neither inertia-based criteria nor qualitative inspection suggests the existence of a stable or naturally separable partitioning of the feature space.

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This instability indicates that the data does not satisfy the structural assumptions underlying k-means clustering. Rather than revealing latent group structure, the algorithm imposes artificial boundaries on a space dominated by continuous variation and correlated features.

These results further reinforce the conclusion drawn from PCA diagnostics: the current feature construction does not support meaningful unsupervised partitioning.

5.5.4 Methodological Implications for Iterative Refinement

The diagnostic analyses conducted in this section highlight the limits of structural modeling under the present feature engineering choices. PCA reveals variance concentration and redundancy without interpretable low-dimensional structure, while clustering fails to identify stable partitions.

Crucially, these outcomes do not represent analytical failure, but rather provide actionable guidance for methodological refinement. They motivate a shift away from further unsupervised modeling toward revisiting feature design, temporal framing, and evaluation criteria.

This diagnostic stopping point exemplifies an iterative, feedback-driven research strategy, where analytical decisions are guided by empirical signals rather than by adherence to predetermined methodological pipelines.

The absence of interpretable low-dimensional structure and stable unsupervised partitioning raises a critical methodological question: how should analytical progress be evaluated when traditional modeling signals fail to provide clear guidance? In such contexts, continuation based solely on methodological momentum risks reinforcing uninformative iterations.

This observation motivates the introduction of explicit evaluation and decision rules, designed to assess iteration value and determine when further modeling is warranted. The formulation of these criteria is the focus of the following section.

5.6 Evaluation, Benchmarking, and Decision Rules

This section addresses a central methodological challenge in exploratory, data-driven research: how to evaluate analytical progress and determine whether further iteration is justified when no ground truth labels or objective success metrics are available. Building on the diagnostic outcomes of Section 5, this section formalizes evaluation criteria, benchmarking principles, and decision rules that govern continuation, revision, or termination of the analytical process.

Rather than treating evaluation as a post hoc validation step, the approach adopted here positions evaluation as a governance mechanism for iterative research, explicitly designed to prevent unproductive methodological escalation.

5.6.1 Challenges of Evaluation in Exploratory Research

In empirical studies based on large-scale web data, evaluation differs fundamentally from supervised learning or controlled experimental settings. Ground truth labels are typically unavailable, and the primary objective is to extract interpretable patterns rather than optimize predictive accuracy.

Standard supervised metrics such as accuracy, precision, or F1-score are therefore inapplicable. Conversely, unsupervised metrics commonly used in clustering or dimensionality reduction, including inertia or silhouette scores, are highly sensitive to modeling assumptions and data scaling. Optimizing such metrics can lead to illusory improvements that do not correspond to meaningful insight.

In this context, evaluation must be framed relative to the research objectives and the stability, interpretability, and empirical relevance of the results, rather than as an absolute measure of performance.

5.6.2 Benchmarking Beyond Model Performance

Benchmarking in this study is not conceived as a competition between models, but as a comparative instrument to guide methodological decisions across iterations. Unlike production-oriented systems where marginal numerical improvements may have operational significance, the present empirical setting prioritizes conceptual clarity and robustness over metric sensitivity.

A distinction is therefore drawn between:

- **High-sensitivity benchmarks**, appropriate for model optimization contexts where small improvements are meaningful;
- **Empirical benchmarks**, used to assess whether an analytical approach produces stable, interpretable, and decision-relevant insights.

In the present study, the latter perspective is adopted. Benchmarking serves to compare analytical strategies and determine whether they add substantive value relative to their complexity and assumptions.

5.6.3 Formal Decision Rule for Iterative Evaluation

To operationalize evaluation and prevent FOMO-driven or waterfall-style continuation, the methodological choices made throughout the use case are formalized into an explicit decision rule governing analytical iterations.

Let D_t denote the dataset at iteration t , and let $X_t \in \mathbb{R}^{N \times d}$ be the standardized feature matrix. Each iteration proposes a modification Δ_t (e.g., feature engineering, grouping strategy, or modeling choice). The decision rule maps (D_t, X_t, Δ_t) to one of three actions:

$$\text{ACTION}_t \in \{\text{CONTINUE}, \text{PIVOT}, \text{STOP}\}.$$

Gate 0: Data Viability Constraint

To ensure statistical reliability of early feedback features, a minimum support constraint is imposed. Let $n_i^{(0)}$ denote the number of user interactions available for application i in the early observation window. A minimum threshold $r_{\min}$ is required:

$$n_i^{(0)} \geq r_{\min}.$$

If more than a fraction α of applications violate this constraint, the iteration is revised:

$$\frac{1}{N} \sum_{i=1}^N \mathbb{I}[n_i^{(0)} < r_{\min}] > \alpha \quad \Rightarrow \quad \text{PIVOT}.$$

Gate 1: Variance Redundancy Diagnostic

Principal Component Analysis is used as a diagnostic tool to assess variance concentration. Let $EV R_t(k)$ denote the cumulative explained variance ratio of the first k components, and define:

$$k^* = \min\{k \mid EV R_t(k) \geq \theta\}, \quad R_t = \frac{k^*}{d}.$$

If variance is dominated by a small number of correlated components and $R_t \leq \rho$, the feature space is considered redundant:

$$R_t \leq \rho \quad \Rightarrow \quad \text{PIVOT.}$$

Gate 2: Partition Feasibility via Clustering Stability

To assess whether the feature space supports meaningful partitioning, k-means clustering is applied as a structural stress test. Stability is evaluated across multiple values of k and random initializations using the Adjusted Rand Index (ARI). Let S_t^* denote the maximum observed stability.

If no stable partition emerges:

$$S_t^* < \sigma \quad \Rightarrow \quad \text{PIVOT.}$$

Gate 3: Distribution-Aware Rule-Based Stratification

When model-based clustering is rejected, grouping is performed using distribution-aware decision rules. Let Y denote a key empirical indicator (e.g., interaction or popularity score), with quartiles Q_1, Q_2, Q_3 and interquartile range $IQR = Q_3 - Q_1$. An upper threshold is defined as:

$$T_{\text{out}} = Q_3 + \lambda \cdot IQR.$$

Applications are stratified according to:

$$\text{Class}(y) = \begin{cases} 0 & y < Q_1, \\ 1 & Q_1 \leq y < Q_2, \\ 2 & Q_2 \leq y < Q_3, \\ 3 & Q_3 \leq y \leq T_{\text{out}}, \\ 4 & y > T_{\text{out}}. \end{cases}$$

This approach preserves interpretability and accommodates heavy-tailed distributions without imposing artificial cluster structure.

Gate 4: Iteration Stopping Criterion

To avoid unproductive iteration, a minimal improvement criterion is defined. Let $V(t)$ denote an iteration value function:

$$V(t) = w_1 S_t^* + w_2 (1 - R_t) + w_3 I_t,$$

where I_t is an interpretability indicator. If:

$$V(t) - V(t-1) < \varepsilon \quad \text{for } m \text{ consecutive iterations,}$$

the analytical process is terminated:

$$\text{STOP.}$$

5.6.4 Application of the Decision Rule to the Use Case

Applying the decision rule to the present study yields the following outcomes. PCA diagnostics reveal variance redundancy and dominance of scale-driven features, while clustering stability analysis fails to identify robust partitions. These signals jointly trigger a pivot away from model-based unsupervised methods.

The rule-based stratification based on quartiles and interquartile range provides an interpretable and empirically grounded alternative, enabling meaningful comparison across popularity levels without overfitting or assumption violation.

5.6.5 Methodological Implications

The evaluation framework presented in this section demonstrates that analytical stopping is a valid and necessary methodological outcome. Rather than indicating failure, stopping reflects informed judgment based on empirical diagnostics.

By formalizing evaluation and decision rules, this approach mitigates the risk of unproductive iteration, aligns analytical effort with empirical signal strength, and provides a transferable framework for governing exploratory research beyond the present case study.

5.7 Descriptive Results

This section reports the empirical observations obtained from the constructed dataset following the methodological process described in Sections 4–6. The results are presented as descriptive statistics and frequency-based observations derived from the applied selection rules and stratification criteria, without introducing causal interpretation or generalization.

5.7.1 Popularity Repartition by Initial Impression

The distribution of final popularity classes is examined with respect to the initial impression classes obtained from early user feedback.

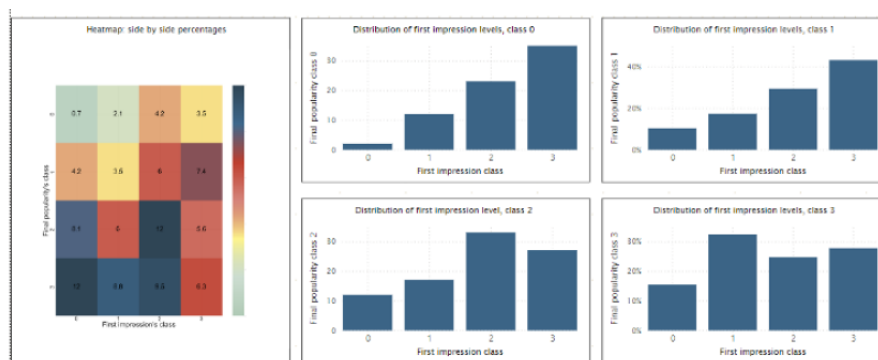
Half of the applications ending up in the lowest final popularity class originated from applications that started with a better initial impression. This indicates that unfavorable final outcomes are not restricted to applications with poor early reception.

Approximately 72% of the applications ending with class 1 popularity originated from applications that initially exhibited good to very good first impressions. This reflects a concentration of intermediate final outcomes among applications that received relatively favorable early feedback.

Around 40% of the applications starting with a class 2 first impression maintained their popularity class over time, indicating partial persistence between initial and final regimes for this subgroup.

Applications reaching the highest popularity class originate from all types of initial impression classes. No exclusive association between high initial impression and highest final popularity is observed in the dataset.

These observations are consistent with findings reported by McIlroy, Ali, and Hassan in their study *“Fresh apps: an empirical study of frequently-updated mobile apps in the Google Play Store”*, which shows that approximately 25% of changes introduced during application updates have negative effects on popularity metrics. This contextualizes the observed transitions by highlighting that application trajectories may evolve independently of early reception.



5.7.2 Repartition of Final Outcomes Within Initial Classes

The analysis of final popularity outcomes within each initial impression class further illustrates the distributional nature of popularity transitions.

Applications starting with low initial impressions are distributed across all final popularity classes, including a limited number of applications reaching high popularity levels.

Applications starting with intermediate initial impressions show higher frequencies of final outcomes in intermediate popularity classes, while still exhibiting dispersion across the full range of final regimes.

Applications starting with strong initial impressions are also dispersed across final popularity classes, without exclusive concentration in the highest popularity regime.

5.7.3 Structural Characteristics of the Observed Distributions

Across all comparisons, popularity-related variables exhibit asymmetric and heavy-tailed distributions. Transitions between initial and final classes do not form discrete or mutually exclusive groupings but instead appear as continuous redistributions across popularity levels.

These structural characteristics are preserved by the rule-based stratification strategy and are consistent with the diagnostic analyses reported in earlier sections.

5.8 Limits of Progression Toward RQ3

The third research question (RQ3) aims to investigate whether user reviews can explain or characterize the mechanisms underlying application popularity patterns. Addressing this question requires the prior identification of stable and interpretable popularity regimes that can be meaningfully compared through textual analysis.

However, the results obtained in this study do not provide sufficient empirical basis to support such a progression.

As shown in the descriptive and evaluative analyses, the relationship between early user feedback and later popularity outcomes does not yield clear, discrete, or stable groupings of applications. Popularity trajectories remain continuous and overlapping across initial impression regimes, and no classification produces groups that can be interpreted as distinct behavioral or success categories.

In the absence of robust or well-defined application classes, proceeding to a review-based analysis using topic modeling techniques such as Latent Dirichlet Allocation (LDA) would introduce an additional layer of assumptions. In particular, applying LDA would implicitly presume that applications can be partitioned into meaningful categories whose textual content reflects underlying causal mechanisms. Given the observed distributional overlap and the lack of separable structure, this assumption cannot be justified within the scope of the current results.

Moreover, even if topic models were to identify recurrent themes in user reviews, these topics could not be reliably interpreted as explanatory factors of popularity patterns. Without a stable mapping between review groups and empirically distinct popularity regimes, any association between discovered topics and observed outcomes would remain ambiguous and potentially spurious.

Consequently, although supervised LDA was executed, this report does not advance to RQ3. This decision reflects a methodological constraint rather than an absence of analytical capabilities. Proceeding without a defensible intermediate structure would risk attributing causal significance to textual patterns that are not grounded in the empirical organization of the data.

This limitation highlights the dependency structure of the research questions and underscores the role of evaluation and stopping criteria in governing methodological progression within empirical research.

PART III

Synthesis — Linking Methodological Perspective and Empirical Study

CHAPTER 6

Synthesis: From Empirical Execution to Methodological Insight

6.1 What the Use Case Demonstrates

The empirical study conducted in this thesis does not demonstrate a causal relationship between first impression and application popularity. Instead, it demonstrates how a predefined research trajectory unfolds when the empirical signal remains weak, continuous, and non-separable.

The use case reflects a situation frequently encountered in data-driven research: extensive data availability, mathematically sound methods, and careful analytical execution coexist with limited structural signal. Despite rigorous feature engineering and diagnostic analysis, the data does not yield clear clusters, stable partitions, or predictive separability.

This absence of discrete structure is not anomalous. It reflects an empirical condition inherent to systems driven by cumulative, overlapping, and partially exogenous dynamics. The study is therefore informative not because of what it establishes empirically, but because of how the research process responds to persistent ambiguity.

6.2 Misalignment Between Research Questions and Empirical Structure

The initial research questions implicitly assume the existence of distinguishable popularity regimes, stable mappings between early user feedback and long-term outcomes, and analyzable groupings that would enable subsequent comparative analysis.

The empirical structure observed in the data contradicts these assumptions. Popularity evolves as a continuous, heavy-tailed process. Early impressions do not induce separable classes, and transitions between initial and final states remain overlapping and non-exclusive.

This reveals a structural mismatch between the formulation of the research questions and the nature of the empirical system. The questions are framed categorically, while the data is fundamentally distributional. This mismatch constitutes a central methodological insight: when ambiguity is structural rather than accidental, analytical refinement alone cannot resolve it.

6.3 Methodological Momentum as an Empirical Phenomenon

In response to this structural ambiguity, the execution of the study exhibits a pattern of methodological momentum. Analytical depth increased despite the absence of reinforcing diagnostic signals. Clustering and classification were combined even after stability analyses failed to identify robust partitions. Dimensionality reduction and unsupervised modeling were explored despite variance redundancy and structural overlap.

These analytical choices were not technically incorrect. Each step conformed to established methodological practice. What becomes visible, however, is the absence of mechanisms that translate diagnostic outcomes into decisive methodological action. In the absence of such mechanisms, continuation becomes the default response to uncertainty.

6.4 Behavioral Forces Shaping Research Under Uncertainty

The persistence of methodological momentum cannot be explained solely by technical considerations. It reflects behavioral forces that shape how researchers act under uncertainty, independently of the specific domain or dataset.

As discussed earlier in the thesis, open-ended research settings activate well-documented cognitive phenomena: fear of missing alternative insights, over-control through exhaustive exploration, and the tendency to justify prior analytical investment. These forces encourage continuation even when empirical signals weaken, not because continuation is justified, but because stopping lacks psychological and structural support.

The use case provides a concrete illustration of these general dynamics. The accumulation of methods, the hesitation to simplify, and the delayed transition toward evaluative reasoning mirror behavioral patterns observed across exploratory scientific practice. These behaviors are systemic, not idiosyncratic, and arise naturally when research structures do not encode explicit decision points.

6.5 What the Case Study Teaches About Linear Research Organization

The empirical trajectory does not demonstrate an absence of iteration. Iterative analysis occurred throughout the study. The limitation lies in how iteration was organized.

The research followed a linear progression because its structure did not integrate mechanisms for reassessment, redirection, or termination in response to diagnostic feedback. In this context, linear organization interacts with behavioral pressure to produce inertia: progression continues even as empirical justification weakens.

This observation clarifies the limitation of waterfall-style research design in exploratory settings. The issue is not rigidity or lack of sophistication, but the absence of structural elements that counteract behavioral tendencies toward uncontrolled continuation.

6.6 Agile Research Logic as a Structural Alternative

The methodological framework developed earlier in the thesis proposes an alternative logic for organizing research under uncertainty. Drawing on agile and iterative principles, this logic emphasizes continuous evaluation, explicit benchmarks, and decision points that guide progression.

Within this perspective, decision rules represent only one possible operational mechanism. Other approaches—such as staged evaluation, comparative baselines, or adaptive reformulation of research questions—can serve similar roles depending on context. The defining feature of the alternative is not the tool itself, but the integration of evaluation into the structure of the research process.

The use case highlights the consequences of operating without such structures. It does not prescribe a unique solution, but illustrates why research organization must explicitly account for both empirical ambiguity and behavioral pressure.

6.7 Transferability Beyond the Use Case

Although grounded in a specific empirical context, the insights derived from this synthesis extend beyond mobile application research. Any data-driven study involving large-scale observational data, unsupervised exploration, ambiguous signal, and complex analytical pipelines is subject to similar dynamics.

The empirical study functions as a mirror. It reflects not the behavior of users or systems, but the behavior of research itself when confronted with uncertainty. By making these dynamics explicit, the synthesis establishes a foundation for rethinking how research is structured, evaluated, and governed across scientific domains.

PART IV

General conclusion

CHAPTER 7

Conclusion

The empirical study examined in this work exposes structural and behavioral constraints that arise when exploratory, data-driven research is executed under linear research strategies. The observed outcomes provide concrete evidence of how research organization, rather than technical capability alone, conditions analytical progress and interpretability.

7.1 Empirical Constraints Revealed by the Case Study

The descriptive and diagnostic analyses conducted on mobile application data show that early user feedback does not induce stable or separable patterns with respect to long-term popularity outcomes. Applications with similar early impressions exhibit divergent trajectories, while highly popular applications originate from a wide range of initial conditions.

Across multiple analytical stages, the data exhibits overlapping distributions, redundant variance, and continuous transitions rather than discrete structure. These properties limit the applicability of unsupervised partitioning and constrain progression toward explanatory modeling.

These empirical characteristics do not contradict the validity of the analysis. Instead, they delineate the boundaries of inference imposed by the data and the chosen representation.

7.2 Consequences of Linear Research Execution

The empirical trajectory demonstrates how, in the absence of intermediate evaluation mechanisms, analytical progression may continue independently of evidential differentiation. When research questions and analytical pipelines are fixed early, empirical ambiguity does not naturally trigger revision or termination.

As a result, analytical complexity can increase without corresponding gains in interpretability. This behavior reflects a structural property of waterfall research execution: continuation is procedural rather than conditional on intermediate empirical signal strength.

The case study illustrates how such dynamics can lead to extensive analysis while leaving core research questions unresolved.

7.3 Behavioral Interpretation of Research Progression

Beyond methodological structure, the observed research trajectory is consistent with cognitive tendencies described in behavioral psychology. Under conditions of uncertainty, decision-making may favor continued exploration over early resolution, particularly when intermediate results remain ambiguous rather than decisively negative.

In the present case, progression toward increasingly complex analytical stages occurred despite weak structural support from earlier diagnostics. This pattern aligns with interpretations discussed earlier in the thesis, where Fear of Missing Out and over-control tendencies encourage exhaustive exploration in search of latent signal.

The empirical study thus demonstrates that research outcomes reflect not only data properties and modeling choices, but also behavioral responses to uncertainty embedded within research processes.

7.4 Methodological Implications

The case study highlights the necessity of treating research strategy as an object of analysis in its own right. In data-driven contexts, methodological rigor extends beyond correctness of implementation to include the ability to evaluate progress, recognize empirical limits, and govern analytical continuation.

The observed constraints support the need for research frameworks that incorporate diagnostic evaluation and decision mechanisms capable of responding to intermediate results. Such mechanisms serve to align analytical effort with empirical signal strength and to mitigate predictable escalation driven by uncertainty.

7.5 Concluding Perspective

Taken together, the empirical outcomes and methodological reflections presented in this work demonstrate that research value is not solely determined by conclusive empirical findings. Equally informative are the constraints revealed when data resists simplification or categorization.

By grounding methodological critique in a concrete empirical case, this work clarifies how linear research execution and behavioral dynamics interact to shape research outcomes. The conclusions drawn here therefore extend beyond the specific application domain and contribute to a broader understanding of how data-driven research can be conducted with greater control, transparency, and interpretive discipline.

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